

Proof. **Chebyshev's Sum Inequality.** Let

$$a_1 \leq a_2 \leq \cdots \leq a_n, \quad b_1 \leq b_2 \leq \cdots \leq b_n$$

be real numbers. We shall prove

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right), \quad ()$$

and describe the equality case.

1. **Base case and degenerate sequences.** If $n = 1$ the inequality reads $a_1 b_1 \geq a_1 b_1$, true with equality.

If all a_i are equal, say $a_i = a$ for all i , then

$$n \sum_{i=1}^n a_i b_i = a \sum_{i=1}^n b_i = \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right),$$

so $()$ holds as an equality. The same argument works when the b_i are all equal. Henceforth we assume $n \geq 2$ and that *not both* sequences are constant.

2. **A convenient notation.** Define

$$S := n \sum_{i=1}^n a_i b_i - \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right).$$

Our goal is to show $S \geq 0$.

3. **Transforming S into a symmetric double sum.**

The product of the two simple sums can be written as

$$\left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right) = \sum_{i=1}^n \sum_{j=1}^n a_i b_j.$$

Likewise

$$n \sum_{i=1}^n a_i b_i = \sum_{i=1}^n \sum_{j=1}^n a_i b_i,$$

since for each fixed i the inner sum over j contributes exactly n copies of the term $a_i b_i$. Hence

$$\begin{aligned}
S &= \sum_{i=1}^n \sum_{j=1}^n a_i b_i - \sum_{i=1}^n \sum_{j=1}^n a_i b_j \\
&= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n (a_i b_i - a_i b_j - a_j b_i + a_j b_j) \\
&= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n (a_i - a_j)(b_i - b_j). \tag{1}
\end{aligned}$$

Equation (1) is the desired double-sum representation of S .

4. **Non-negativity of each summand.** Take any ordered pair (i, j) with $i < j$. Because the sequences are non-decreasing,

$$a_i \leq a_j, \quad b_i \leq b_j,$$

so

$$a_i - a_j \leq 0, \quad b_i - b_j \leq 0,$$

and consequently

$$(a_i - a_j)(b_i - b_j) \geq 0.$$

When $i > j$ the same product appears (the double sum in (1) is symmetric); therefore $(a_i - a_j)(b_i - b_j) \geq 0$ for *every* pair (i, j) .

From (1) we obtain

$$S = \frac{1}{2} \sum_{i,j} (a_i - a_j)(b_i - b_j) \geq 0.$$

Re-substituting the definition of S yields precisely inequality ().

5. **Equality condition.** Equality $S = 0$ occurs iff each summand in (1) vanishes, i.e.

$$(a_i - a_j)(b_i - b_j) = 0 \quad \text{for all } 1 \leq i, j \leq n.$$

Thus for every unordered pair $\{i, j\}$ we must have either $a_i = a_j$ or $b_i = b_j$. The only way this can happen for all pairs simultaneously is that one of the two sequences is constant:

- either $a_1 = a_2 = \cdots = a_n$,

- or $b_1 = b_2 = \cdots = b_n$.

The case $n = 1$ trivially satisfies both alternatives. Hence equality in Chebyshev's inequality holds exactly when one of the sequences is constant.

6. **Concluding remark.** The same inequality follows from the rearrangement inequality, which states that for two similarly ordered sequences the sum $\sum a_i b_{\sigma(i)}$ is maximal when the permutation σ preserves the ordering (the identity permutation). Taking σ to be the identity gives Chebyshev's sum inequality directly.

□